FINAL UVM VERIFICATION PROJECT

Complex AHB-Lite Multi-Function Peripheral

1. Objective

Develop a complete UVM verification solution for the provided 32-bit AHB-Lite peripheral. The DUT is provided. The trainee must study the specification, produce the verification plan and derive the verification architecture, test strategy, scoreboard model, coverage and regression.

2. DUT Interface

Signal / Group	Width	Purpose
HCLK / HRESETn	1	Clock / active-low reset
HSEL	1	Slave select
HADDR	32	Address
HTRANS	2	Transfer type
HWRITE	1	Read/write
HSIZE	3	Transfer size
HBURST	3	Burst type
HPROT	4	Protection attributes
HWDATA / HRDATA	32	Write / read data
HREADY / HREADYOUT	1	Pipelined transfer completion
HRESP	1	Error response
IRQ	1	Peripheral interrupt

3. DUT Functional Blocks

- 2 KiB byte-addressable RAM.
- Register bank.
- 16-entry x 32-bit FIFO.
- Programmable timer.
- Interrupt controller.
- Programmable/random wait states.
- AHB-Lite burst and transfer-size handling.
- Illegal-access and error-response logic.

4. Address / Register Map

Address	Block / Register	Description
0x0000-0x07FF	RAM	2 KiB memory
0x1000	CTRL	Peripheral control
0x1004	STATUS	Read-only status
0x1008	INT_EN	Interrupt enable
0x100C	INT_STATUS	W1C interrupt status
0x1010	TIMER_LOAD	Timer preload
0x1014	TIMER_CTRL	Timer control
0x1018	TIMER_VALUE	Read-only timer value
0x1020	FIFO_DATA	Push / pop
0x1024	FIFO_STATUS	Read-only FIFO status
0x1028	SCRATCH	General R/W register
0x102C	WAIT_CFG	Maximum inserted wait states

5. Minimum DUT Behavior

- Support byte, half-word and word transfers.
- Support SINGLE, INCR, WRAP4 and INCR4 burst encodings used by the supplied DUT.
- Address/control and data belong to different AHB phases.
- The slave may insert 0-3 wait states.
- Misaligned accesses, unsupported transfer sizes/bursts, unmapped addresses and illegal register/FIFO accesses return HRESP error.
- FIFO overflow/underflow are error conditions.
- Timer supports one-shot and periodic operation.
- Interrupt status is W1C.

- Reset clears control state, timer, FIFO and RAM.

6. Project Expectation

- Do not begin with UVM code. Begin with DUT understanding and a verification plan.
- Define verification features, risks, testbench architecture, stimulus strategy, checking strategy, assertions, coverage, tests, regression and exit criteria.
- The monitor/checking path must be independent of the driver.
- The environment must be self-checking.
- Directed, constrained-random, negative and stress verification are expected.
- Technical questions and unclear cases are to be raised during mentor reviews.

7. Submission

- Complete verification repository.
- Verification plan.
- UVM testbench.
- Regression results.
- Functional coverage report.
- Final verification report.
- Bug reports for defects found.